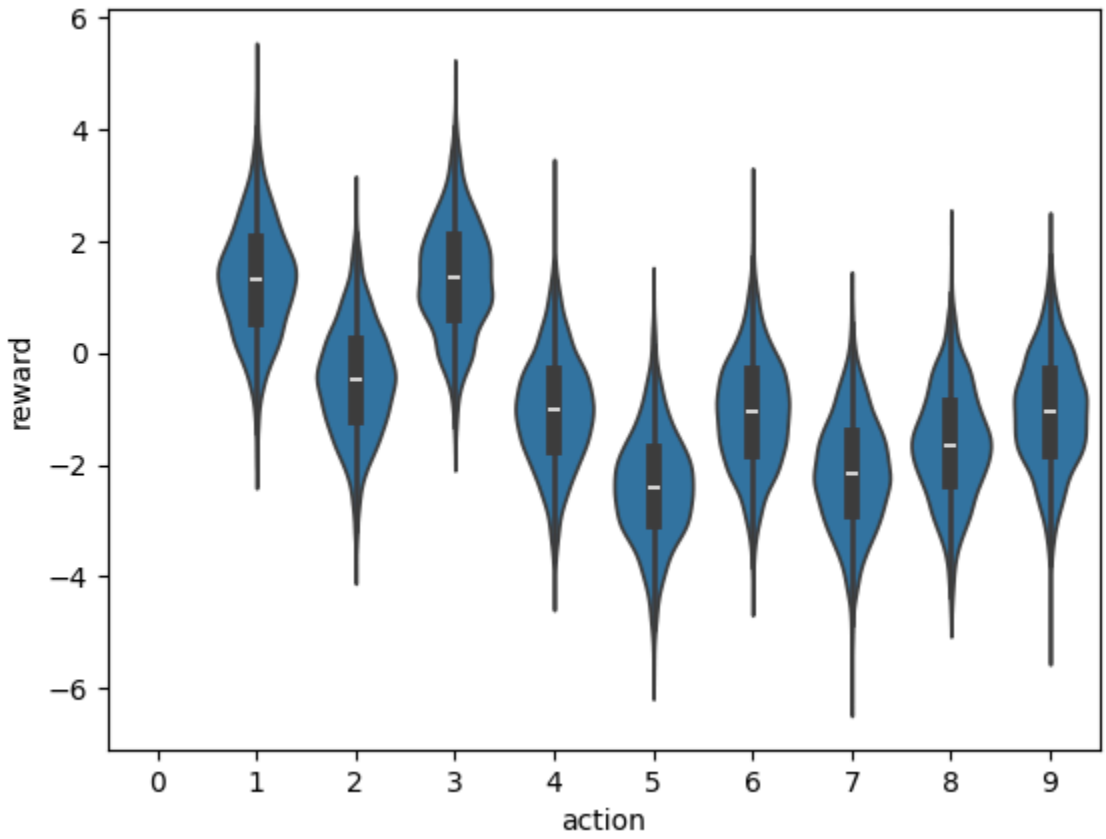
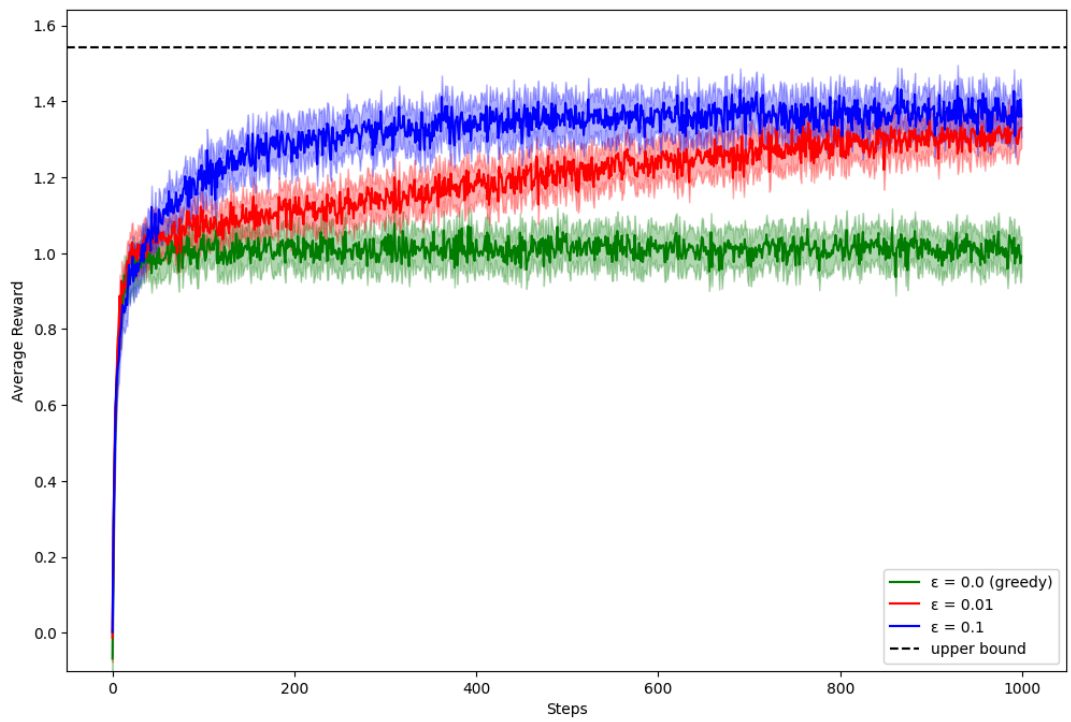
Collaborated on homework with Sam Milhaven

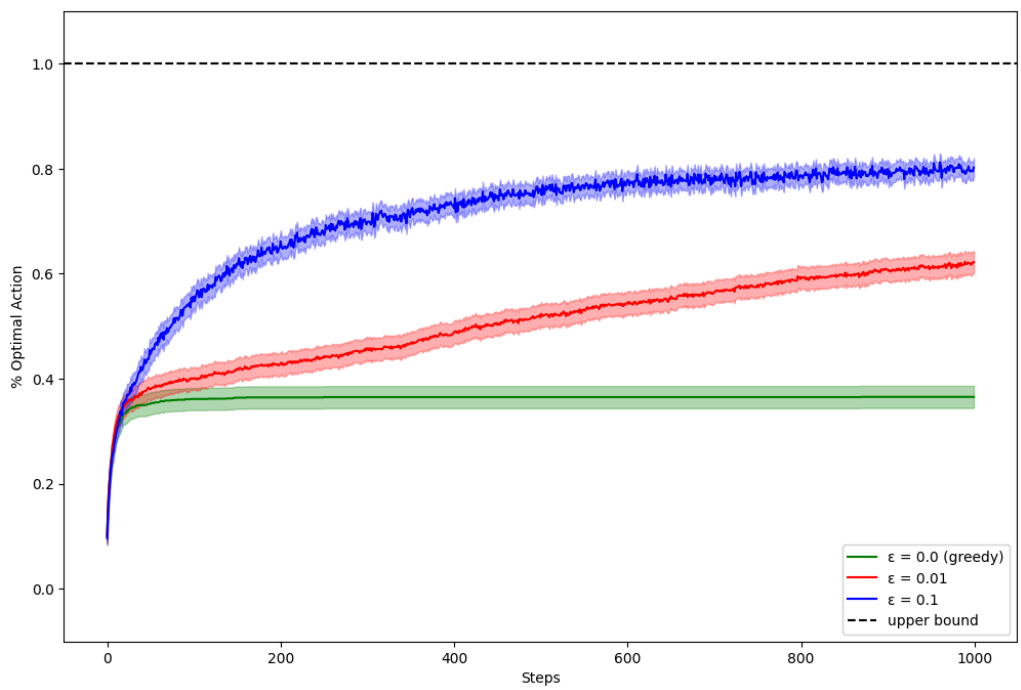
Q1.) Plot



Q2.) Plot 1



Plot 2



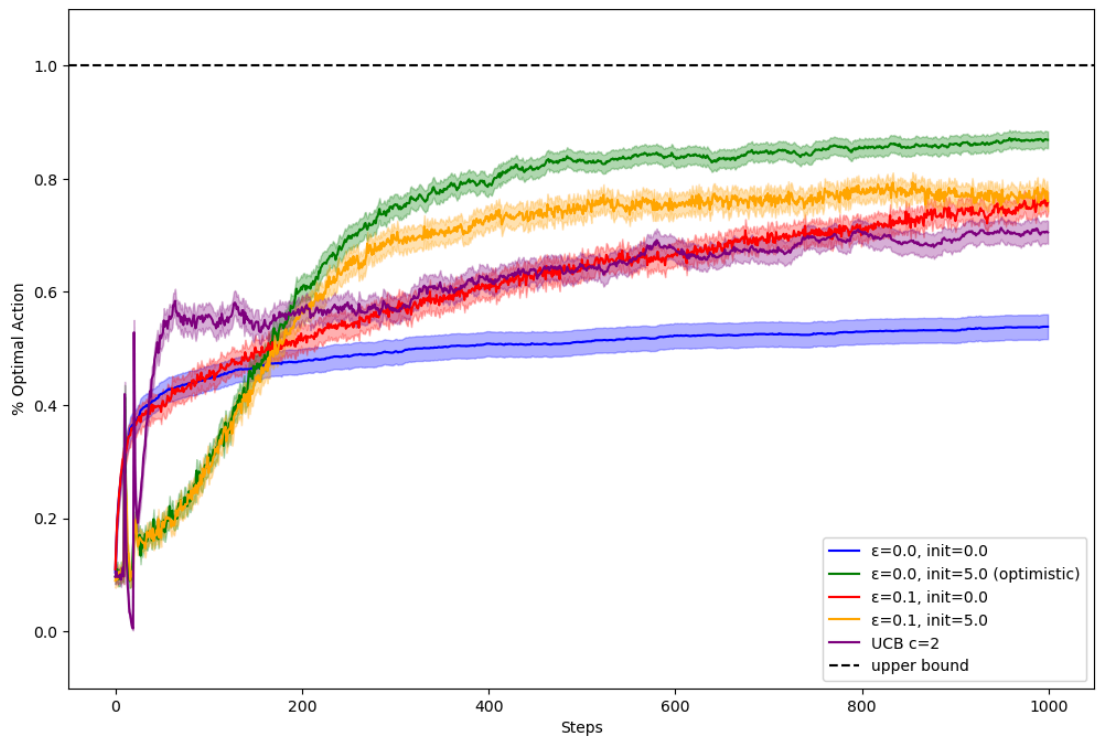
Written: What are the average rewards that the algorithm converges to using different ε values? Why does the algorithm converge to different average rewards for different ε? Explain briefly.

Using ε = 0.0 it will average out to 1 for my average rewards. For 0.01 the average starts out at 1 but continues to rise as the longer it goes on so by the end of my steps it hits about 1.3. Then if my ε is 0.1 I will average out to 1.4 about a fifth of a way into my experiment. These all

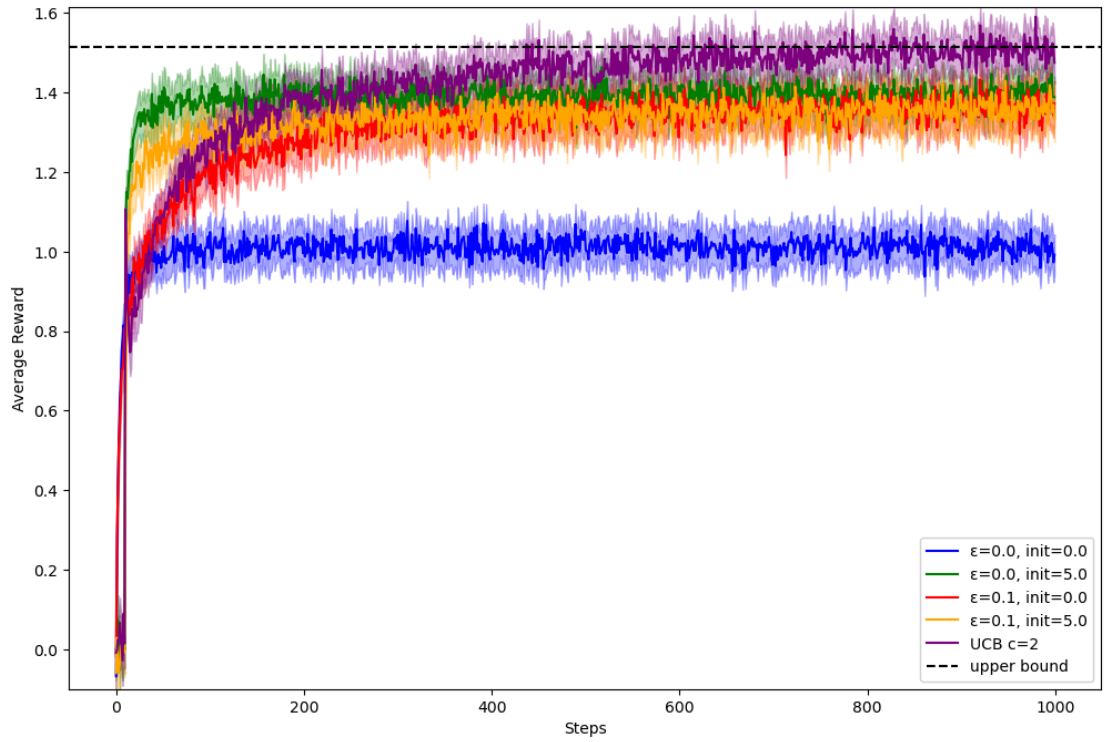
converge to different average rewards based on our greedy usage. If we are very greedy, we will never gain new rewards, and will stagnate very fast. We will not be exploring enough if we use a small ε., we will always get more rewards but it'll be slow. If we start with a larger ε we will get new rewards fast, but will start to peter out because we're not prioritizing exploitation enough.

Q3.)

Plot 1:



Plot 2:



Written: We have seen the spike for optimistic initialization in class (Figure 2.3, Figure 2.4 in RL2e). Observe that UCB also produces spikes in the very beginning in both the two reproduced figures. Explain in your own words why the spikes appear (both the sharp increase and sharp decrease). Analyze and use your experimental data as further empirical evidence to back your reasoning.

These spikes appear because we are telling our agents to immediately explore and for longer in the beginning. The first spike is because we find so many new paths and thus might stumble into new rewards right away, then the decrease is because we find more paths that are bad and don't work out for our agent. The longer it goes after those spikes the better it will start to perform. By looking at my plot I am able to see that my optimistic ε-greedy gets the exploration out of the way and then prioritizes more on exploitation.

Q4.)

Q4.) α_n not const Q_n weight an

what is the weighting on each prior reward

for the general case in terms of step size

$$Q_{n+1} = Q_n + \alpha_n [R_n - Q_n]$$

$$Q_n + \alpha_n R_n - \alpha_n Q_n$$

$$(1-\alpha_n) Q_n + \alpha_n R_n$$

$$(1-\alpha_n)(1-\alpha_{n-1}) Q_{n-1} + \alpha_{n-1} R_{n-1}) + \alpha_n R_n$$

$$\prod_{i=1}^{n} (1-\alpha_i) Q_1 + \sum_{i=1}^{n} \left[\alpha_i \prod_{j=i+1}^{n} (1-\alpha_j) \right] R_i$$

$$R_i = \alpha_i \prod_{j=i+1}^{n} (1-\alpha_j)$$

Q5.)
a.)

a.) The sample-average is unbiased because since $Q_1 = \frac{R_1 + \ldots + R_n}{n}$ and R_i is a sample from the reward distribution of arm a

$$E[Q_1] = E\left[\frac{1}{n}\sum_{i=1}^{n} R_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[R_i] = \frac{1}{n} \cdot n \cdot q_* = q_*$$

so the expected value of the estimate equals the true value.

b.) It is biased. Since the initial estimate Q1 = 0 and the exponential recency-weighted update rule, the estimate at the timestep n depends on the initialization. Unless qstar equaled 0 our expected output for Q1 would be 0 which would not equal qstar. Thus this would introduce bias. The bias would persist for finite time because the exponential weighting gives non-zero weight to all past estimates including Q1

c.)

c.)

$$Q_{n+1} = Q_n + \alpha(R_n - Q_n) = (1-\alpha)Q_n + \alpha R_n$$

$$Q_2 = (1-\alpha)Q_1 + \alpha R_1$$

$$Q_3 = (1-\alpha)Q_2 + \alpha R_2 = (1-\alpha)^2 Q_1 + \alpha(1-\alpha)R_1 + \alpha R_2$$

$$Q_n = (1-\alpha)^{n-1} Q_1 + \alpha \sum_{i=1}^{n-1} (1-\alpha)^{n-1-i} R_i$$

$$E[Q_n] = (1-\alpha)^{n-1} Q_1 + \alpha \sum_{i=1}^{n-1} (1-\alpha)^{n-1-i} E[R_i]$$

$$E[R_i] = q_* \qquad \downarrow \; = \frac{1-(1-\alpha)^{n-1}}{\alpha}$$

$$E[Q_n] = (1-\alpha)^{n-1} Q_1 + \alpha q_* \left[1-(1-\alpha)^{n-1}\right]$$

$$E[Q_n] - q_* = (1-\alpha)^{n-1}(Q_1 - q_*)$$

d.)

d.) show $n \to \infty$

$$Q_{n+1} = Q_n + \alpha(R_n - Q_n) = (1-\alpha)Q_n + \alpha R_n$$

$$Q_n = (1-\alpha)^{n-1} Q_1 + \alpha \sum_{i=1}^{n-1} (1-\alpha)^{n-1-i}$$

$$E[Q_n] = (1-\alpha)^{n-1} Q_1 + \alpha q_* \sum_{i=1}^{n-1} (1-\alpha)^{n-1-i}$$

$$E[Q_n] - (1-\alpha)^{n-1} Q_1 = \frac{1-(1-\alpha)^{n-1}}{\alpha}$$

Therefore

$$E[Q_n] = (1-\alpha)^{n-1} Q_1 + \left[1-(1-\alpha)^{n-1}\right] q_*$$

$n \to \infty$ since $0 < \alpha < 1$

$$\lim_{n \to \infty} E[Q_n] = 0 \cdot Q_1 + 1 \cdot q_* = q_*$$

so Q_n is asymptotically unbiased

e.) We should expect it to be biased in general because it is non-uniform weighting, has persistent influence on the initialization, and the weights don't sum to 1 properly for finite n. Therefore the bias is adaptive and the exponential average tracks non-stationary reward distributions better by forgetting old data. However this comes at the cost of being finite-sample bias.